

Omar Elmetwalli

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Education

Zewail City of Science and Technology

Expected June 2027

B.Sc. Aerospace Engineering, Minor in Communications and Information Engineering

Cumulative GPA: 3.303 / 4.00

Research Experience

ML-Augmented Turbulence Modeling — Manuscript Co-Author

In Progress

Collaborators: Dr. Mahmoud Ayyad (Stevens Institute of Technology), Ahmed Hamada (UT Dallas)

- Developing a physics-informed 3D U-Net for wind-farm LES data, enforcing momentum-deficit conservation, actuator-line-model body-force conditioning, and vorticity transport constraints
- Independently building a 2D adaptation of the modeling pipeline (Leray projection, adaptive correction module) to prototype the physics-constraint layers

Self-Steering Cell Biology — Research Collaborator

Ongoing

With Dr. Mahmoud Abdelgalil (University at Buffalo) and Mahmoud Mashal

- Investigating self-generated gradient dynamics using Keller–Segel models, at the ground floor of a new research direction in the group

GPR Airfoil Surrogate Modeling — Independent Project

- Built an 81-case OpenFOAM RANS sweep (Spalart–Allmaras) and trained a Gaussian Process Regression surrogate; achieved leave-one-case-out cross-validated $R^2 > 0.997$

Leadership

Founder & Chair — AIAA Zewail City Student Branch

Egypt's first AIAA-affiliated chapter

- Built the branch from zero to 60+ members with a full departmental structure (Technical, Marketing & Media, PR & Outreach, HR & Operations, Fundraising)
- Organized “*D’Alembert’s Paradox: A Tale of Two Schools*,” a branch panel featuring Prof. Haitham Taha (UC Irvine), sponsored by Labtronic
- Lead CFD/Aerodynamics, CubeSat, and Design/Build/Fly project teams

Industry & Practical Experience

R&D Engineer, Turbomachinery — [Company Name]

Current

- Contributing to R&D work heading toward a patent filing

Digital Twin Developer (Independent) — Elsewedy Electrometer

- Built a physics-informed digital twin proof-of-concept for predictive maintenance

Process Automation Intern — Anasia Group

Exposure to Valmet and Endress+Hauser instrumentation and control systems

CubeSat Intern — Aqmmar Space Technologies

Aerospace Education — SpacePoint UAE

Aircraft Maintenance Training — AEROSKY Academy

Awards

3rd Place Worldwide, eVTOL Design Category — ROTORTON

2024

Technical Skills

CFD: OpenFOAM, ANSYS Fluent | **Programming:** Python (PyTorch), MATLAB/Simulink, C++ | **CAD:** SolidWorks, Autodesk Inventor | **Typesetting:** L^AT_EX